

Introduction to Information Security

Final Project

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Instructions

Overview

In teams of three, design, implement, and evaluate a secure digital system for a real-world application relevant to emerging markets (e.g., mobile payments, digital identity, secure messaging, health data, or IoT mobility systems).

Key Dates

March 24, 2026, during class: Group selection and project kick-off

April 22, 2026, 11:59PM: Final submission of deliverables

April 23, 2026, during class: Live demo (5mins) and defense (3mins)

Learning Objectives

- Design secure system architectures with clear trust boundaries
- Apply cryptographic primitives correctly (encryption, signatures, hashing)
- Build secure APIs and implement authentication mechanisms
- Conduct threat modeling and adversarial analysis
- Demonstrate attacks and implement effective defenses

Deliverables

1. System Architecture Diagram (with trust boundaries)
2. Threat Model (assets, adversaries, risks)
3. Secure Protocol Design (authentication, key exchange, replay protection)
4. Working Prototype (in C or python, including secure communication and storage)
5. Attack & Defense Demo (break and fix your system)

6. Final Report (max 5 pages)
7. Live Demo + Oral Defense

Requirements

- Must implement encryption, authentication, and secure key handling
- No plaintext communication
- No hardcoded secrets
- No reliance on black-box security services (e.g., Firebase Auth)
- All design decisions must be justified

Timeline (4 weeks)

- **Week 1:** Problem definition, system architecture, and initial threat model
- **Week 2:** Refined threat model and secure protocol design (authentication, key exchange, replay protection)
- **Week 3:** Implementation of core system (secure communication, encryption, authentication)
- **Week 4:** Attack demonstration, defenses, testing, and final demo + report

Evaluation

- Architecture (20%)
- Threat Model (15%)
- Protocol Design (20%)
- Implementation (20%)
- Attack & Defense (15%)
- Clarity & Presentation (10%)

Note

This project emphasizes real-world security thinking. Expect to iterate, make mistakes, and refine your design under adversarial conditions.
